Project Charter: Cookie Cats Gate Placement A/B Test Analysis

Prepared by: Palak Gupta | Date: 15 October 2026 | Status: Final (portfolio case study)

Purpose

Decide whether the first gate in the mobile game Cookie Cats should move from level 30 to level 40, using results from a completed A/B test of 90,188 players, and deliver a clear go or no-go recommendation.

Objectives

1. Determine whether gate_40 changes 1-day and 7-day player retention compared with gate_30, at 95% confidence.
2. Check whether the number of game rounds played changes between the two versions.
3. Find which player groups are affected most.
4. Estimate the business impact per 100,000 installs, with every assumption stated.
5. Deliver the recommendation in a one-page decision memo for non-technical readers.

Scope

In scope

- Data quality checks (duplicates, group balance) and outlier handling
- Significance testing with three methods: Z-test, bootstrap and Bayesian
- Correction for multiple tests, power analysis and player-group analysis
- Business impact estimate, charts and decision memo
- Project documents and a public write-up on GitHub

Out of scope

- Running or redesigning the live experiment
- Revenue and monetisation modelling, because no revenue data is available
- Retention beyond 7 days
- Designing new gate variants

Success criteria

Criterion	How we measure it
Clear statistical answer on 7-day retention	Difference, 95% confidence interval and p-value, confirmed by at least two methods
Data trusted	Zero duplicate players, group balance checked, outlier handled and documented
Result survives scrutiny	Still significant after correcting for multiple tests

Criterion	How we measure it
Decision-ready output	One-page memo with a recommendation, risks and next steps
Business impact understood	Players lost per 100,000 installs, with a low and high case
Work is reproducible	Notebook and write-up published on GitHub

Deliverables

- Analysis notebook (Google Colab)
- Three charts: retention comparison, player groups by rounds played, business impact
- Decision memo (PDF)
- Project charter, stakeholder and RACI table, risk register, timeline
- README on GitHub

Planned timeline

Phase	What happens	Planned duration
1. Setup and data checks	Load data, check duplicates, group balance and outliers	Day 1
2. Core analysis	Retention rates, Z-test, bootstrap, game rounds test, power analysis	Days 1 to 2
3. Deeper analysis	Player groups, multiple-test correction, Bayesian analysis	Days 2 to 3
4. Business impact and memo	Impact estimate, charts, decision memo	Days 3 to 4
5. Project documents and publishing	Charter, RACI, risk register, timeline, GitHub write-up	Days 4 to 5

Stakeholders

Role	Interest	Involvement
Product lead	Makes the launch decision	Receives the memo and recommendation
Game design lead	Owns gate placement and player experience	Reviews findings on player groups
Engineering lead	Would ship any change	Informed of the decision
Data analyst (Palak Gupta)	Project owner	Delivers the analysis and recommendation

These roles are illustrative. The dataset is public (Kaggle) and no real company team was involved. The detailed RACI table follows in a separate document.

Assumptions and constraints

- Players were assigned to gate_30 or gate_40 at random by the original experiment.
- Only 1-day and 7-day retention and rounds played are available.
- The value per retained player (Rs 50) is an assumption for illustration, not a measured figure.
- The analysis is retrospective: the test had already finished before this project began.

Top risks

- A group balance check was flagged (p = 0.0086), although the size gap is small.
- Long-term effects beyond 7 days are unknown.
- The rounds-played result is borderline (p = 0.051), so it cannot show that no difference exists.

The full risk register with owners and mitigation follows in a separate document.